



(12) **United States Plant Patent**
Shaw et al.

(10) **Patent No.:** **US PP19,767 P2**
(45) **Date of Patent:** **Feb. 24, 2009**

(54) **STRAWBERRY PLANT NAMED**
'MONTEREY'

(50) Latin Name: *Fragaria*×*ananassa*
Varietal Denomination: **Monterey**

(75) Inventors: **Douglas V. Shaw**, Davis, CA (US); **Kirk D. Larson**, Irvine, CA (US)

(73) Assignee: **The Regents of the University of California**, Oakland, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **12/011,256**

(22) Filed: **Jan. 25, 2008**

(51) **Int. Cl.**
A01H 5/00 (2006.01)

(52) **U.S. Cl.** **Plt./209**

(58) **Field of Classification Search** Plt./209
See application file for complete search history.

Primary Examiner—Annette H Para

(74) *Attorney, Agent, or Firm*—Townsend and Townsend and Crew LLP

(57) **ABSTRACT**

This invention relates to a new and distinctive day-neutral type of strawberry designated as 'Monterey'. 'Monterey' is a day-neutral (everbearing) cultivar similar to 'Diamante' (U.S. Plant Pat. No. 13,079) but with higher yield and better quality fruit, better disease resistance and better flavor. It is similar to 'Albion' (U.S. Plant Pat. No. 16,228) for fruit quality but with higher yield, and larger fruit.

3 Drawing Sheets

1

Genus and species: The strawberry cultivar of this invention is botanically identified as *Fragaria*×*ananassa* Duch.

Variety denomination: The variety denomination is 'Monterey'.

BACKGROUND OF THE INVENTION

This invention relates to a new and distinctive day-neutral type cultivar designated as 'Monterey', which resulted from a cross performed in 2001 between the cultivar 'Albion' (U.S. Plant Pat. No. 16,228) and advanced selection Cal 97.85-6. 'Monterey' was first fruited at the University of California Wolfskill Experimental Orchard, near Winters, Calif. in 2002, where it was selected, originally designated Cal 1.132-3, and propagated asexually by runners. Following selection and during testing, the plant of this selection was designated 'CN222' and, later for introduction into commerce, 'Monterey'. Asexual propagules from this original source have been tested at the Watsonville Strawberry Research Facility, the South Coast Research and Extension Center, and to a limited extent in grower fields starting in 2005.

BRIEF SUMMARY OF THE INVENTION

'Monterey' is a day-neutral (everbearing) cultivar similar to 'Diamante' (U.S. Plant Pat. No. 13,079) but with higher yield and better quality fruit, better disease resistance and better flavor; it is similar to 'Albion' (U.S. Plant Pat. No. 16,228) for fruit quality but with higher yield, and larger fruit.

BRIEF DESCRIPTION OF THE DRAWINGS

The figures depict various characteristics of the 'Monterey' cultivar.

FIG. 1 shows the general flowering and fruiting characteristics of the plant in a field planting.

FIG. 2 shows a typical leaf at mid-season.

FIG. 3 shows representative mid-season fruit.

2

DETAILED DESCRIPTION OF THE INVENTION

'Monterey' is typical of day-neutral strawberry cultivars and produces fruit regardless of day length when treated appropriately in arid, subtropical climates. 'Monterey' is moderate to weak in expressing the day-neutral character, being comparable in flowering response to 'Diamante' (U.S. Plant Pat. No. 10,435) and 'Albion' (U.S. Plant Pat. No. 16,228), and less so than 'Fern' (U.S. Plant Pat. No. 5,267) or 'Irvine' (U.S. Plant Pat. No. 7,172). The production pattern for 'Monterey' is similar to that for 'Albion'. 'Monterey' will be of special interest for winter plantings and in summer plantings where 'Diamante' and 'Albion' have been successful.

Plants and foliage: Fruiting plants of 'Monterey' are similar in morphology to 'Diamante' and 'Albion' although much larger and more erect. 'Monterey' plants are substantially more open and erect than 'Aromas' plants. Comparative statistics for foliar characters near mid-season are given for 'Monterey' and the three comparison cultivars in Table 1. Individual leaflets for 'Monterey' are similar in shape and size to the comparison cultivars, but are somewhat less rounded. Leaves (including petioles) for 'Monterey' are longer than those for the comparison cultivars, mostly due to greater petiole length. Petioles are generally thicker than those of the comparison cultivars and tend to have heavy pubescence. The adaxial (upper) and abaxial (lower) surfaces of leaves for 'Monterey' are similar in color to the comparison cultivars at mid season. Leaves of 'Monterey' have consistently greater concavity than 'Aromas', 'Diamante', and 'Albion'.

Disease and pest reaction: 'Monterey' is moderately susceptible to powdery mildew (*Sphaerotheca macularis*), and is moderately resistant to Anthracnose crown rot (*Colletotrichum acutatum*) and *Verticillium* wilt (*Verticillium dahliae*). It is moderately susceptible to *Phytophthora* crown rot (*Phytophthora cactorum*) and common leaf spot (*Ramularia tulasnei*) (Table 3). When treated properly, it has tolerance to two-spotted spider

mites (*Tetranychus urticae*) equal to that for the comparison cultivars. ‘Monterey’ is tolerant to strawberry viruses encountered in California.

TABLE 1

Foliar and plant characteristics for ‘Monterey’, ‘Aromas’, ‘Diamante’, and ‘Albion’.				
Foliar Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
Plant height (mm)				
mean	272	220	223	311
range	240-300	190-240	170-290	260-380
Plant spread (mm)				
mean	323	316	295	343
range	300-360	265-385	270-315	315-395
Mid-tier leaflet Length (mm)				
mean	79	78	70	78
range	70-90	60-90	60-80	70-100
Width (mm)				
mean	74	77	68	73
range	70-80	55-90	60-80	60-100
Mid-tier leaf Length (mm)				
mean	113	99	99	111
range	100-120	80-120	90-110	100-140
Width (mm)				
mean	135	134	122	140
range	120-150	90-150	105-135	120-150
Leaf components				
Petiole length (mm)				
mean	174	114	122	212
range	140-210	100-130	95-180	200-230
Petiole diameter (mm)				
mean	4.5	5.2	4.9	5.1
range	4-6	4-7	4-6	4-6
Petiolule length (mm)				
mean	6.6	5.2	6.7	7.4
range	4.3-7.5	4.0-7.6	5.0-8.0	7-8
#	3	3	3	3
leaflets/leaf				
Leaf convexity	some flat, most slight concave	some flat, most slight concave	some flat, most slight concave	Mostly concave
Serrations				
number/leaf	19.9	20.2	23.3	21.4
range	16-24	16-24	21-27	19-24
shape	rounded to semi-pointed	rounded to semi-pointed	semi-pointed	semi-pointed
Leaf pubescence	moderate	moderate- heavy	moderate	heavy
Petiole pubescence	Moderate- heavy	heavy	heavy	moderate- heavy
density				
direction	perpen- dicular	perpen- dicular	perpen- dicular	perpendicular

TABLE 1-continued

Foliar and plant characteristics for ‘Monterey’, ‘Aromas’, ‘Diamante’, and ‘Albion’.				
Foliar Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
Petiole color (Munsell)	5 GY 8/8	7.5 GY 9/4	5 GY 8/8	5 GY 8/8
Stipule length (mm)				
mean	34.2	31.6	32.5	34.7
range	30-39	22-36	24-37	31-38
Stipule color				
core	7.5 GY 8/7	7.5 GY 8/7	5 GY 8/7	5 GY 7/10
margins	2.5 GY 9/3	5 GY 6/8	5 GY 6/8	2.5 GY 7/10
Stolon base diameter (mm)	3.0	3.2	3.0	3.0
Stolons per nursery mother plant	33.0	29.0	26.9	28.6
Venation pattern	pinnate	pinnate	pinnate	pinnate
color	2.5 GY 5/5	10 GY 5/5	2.5 GY 6/8	10 Y 7/9

Flowering, fruiting, fruit, and production characteristics: ‘Monterey’ is similar to other California day-neutral cultivars (e.g. ‘Diamante’ and ‘Albion’) in that it will flower independently of day length, given appropriate temperature and horticultural conditions. Comparative statistics for flower and fruit characters near mid-season are given for ‘Monterey’ and the three comparison cultivars in Table 4. The primary flowers for ‘Monterey’ are similar in size to the comparison cultivars with a calyx that is distinctly larger than the corolla on primary fruit. The sepals are similar in length and shape to those of the comparison cultivars, but are smaller relative to the corolla than for ‘Albion’. The calyx for ‘Monterey’ varies in position but is usually more reflexed than for ‘Aromas’ or ‘Diamante’, similar to ‘Albion’. The fruit shape for ‘Monterey’ can vary but is typically a short and either symmetrical or slightly flattened conic. It is easily distinguished by fruit shape from ‘Aromas’ (shortened and rounded conic), ‘Diamante’ (usually a flat conic) or ‘Albion’ (long conic). ‘Monterey’ usually has a greater proportion of symmetrical fruit than the comparison cultivars. External fruit color for ‘Monterey’ is similar to ‘Aromas’ or ‘Albion’, but distinctly darker than for ‘Diamante’. Internal color is somewhat darker with greater red pigment than for the comparison cultivars (Table 2). Achenes vary from yellow to dark red, but are frequently more yellow than the comparison cultivars, and are even with the fruit surface or slightly indented.

‘Monterey’ has been tested under a variety of cultural regimes, and optimal performance is obtained when nursery treatments and nutritional programs similar to those for ‘Albion’, ‘Diamante’, and ‘Aromas’ are used. In general, ‘Monterey’ is more vigorous than the comparison cultivars and is less sensitive to low chilling. ‘Monterey’ retains excellent fruit quality in summer planting systems.

When treated with appropriate planting regimes, ‘Monterey’ has larger fruit and produces greater individual-plant yield than any of the comparison cultivars (Table 5). ‘Monterey’ has a similar production pattern to ‘Albion’ with most cultural treatments, although it is substantially more adapted to early-season winter planting. Commercial

appearance ratings have been similar to or better than those for all of the comparison cultivars, especially ‘Aromas’. These superior appearance scores translate directly into a larger fraction of marketable fruit than is produced by the comparison cultivars. Fruit for ‘Monterey’ is substantially firmer than fruit from ‘Aromas’, similar in firmness to the other comparison cultivars. Subjectively, ‘Monterey’ has outstanding flavor with a distinct sweet aftertaste. The fruit will be exceptional for both fresh market and processing, and will be useful for home garden purposes.

TABLE 2				
Foliar and fruit color characteristics for ‘Monterey’ and three comparison cultivars.				
Color Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
Leaf color (CIELAB) Adaxial				
L*				
mean	35.1	34.8	34.7	35.4
range	32.7-37.7	32.6-36.8	32.8-36.7	31.3-37.3
a*				
mean	-10.6	-10.4	-9.8	-11.0
range	-8.2--14.0	-8.7--11.9	-9.4--11.3	-9.5--12.2
b*				
mean	13.8	13.8	12.8	14.5
range	11.2-18.1	12.2-16.6	10.7-15.6	12.6-16.3
Munsell	7.5 GY 4/4	5 GY 4/3	5 GY 4/3	5 GY 4/3
Abaxial				
L*				
mean	52.4	51.1	50.6	49.2
range	50.6-54.1	49.7-52.2	43.7-53.1	46.7-52.4
a*				
mean	-11.6	-12.8	-12.4	-12.6
range	-10.7--13.6	-11.6--14.9	-8.6--11.4	-11.8--13.2
b*				
mean	17.3	19.5	17.2	18.4
range	14.3-23.2	15.3-23.5	14.5-19.6	16.4-21.8
Munsell	10 GY 7/8	7.5 GY 6/8	7.5 GY 8/7	5 GY 5/6
Fruit color (CIELAB) External				
L*				
mean	34.2	40.8	36.5	35.0
range	31.2-38.3	35.5-45.4	32.8-40.1	32.0-38.4
a*				
mean	33.9	36.7	33.3	36.1
range	31.5-38.6	35.6-40.2	28.3-36.2	32.2-40.2
b*				
mean	14.1	21.2	17.6	16.3
range	9.1-16.5	18.8-25.7	12.2-24.9	13.3-19.5
Munsell	2.5 R 4/10	5 R 5/13	5 R 3/7	5 R 3/7
Internal				
L*				
mean	61.6	65.6	57.9	48.8
range	59.5-67.7	58.8-67.2	43.3-62.9	30.6-56.0
a*				
mean	14.7	5.6	19.0	31.7
range	7.6-19.2	3.0-9.5	7.9-27.7	26.3-36.8

TABLE 2-continued				
Foliar and fruit color characteristics for ‘Monterey’ and three comparison cultivars.				
Color Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
b*				
mean	20.2	15.8	21.0	28.4
range	16.1-22.5	14.5-18.2	13.2-27.2	24.1-33.0
Munsell	5 R 6/11	10 R 7/9	7.5 R 4/11	7.5 R 6/12
Achene color	7.5 R 4/11	7.5 R 4/11	10 R 5/6	7.5 Y 7/9
Munsell				

*CIELAB is the abbreviation of the international color system known as “Commission Internationale De L’Eclairage” 1978. For recommendations concerning uniform color spaces, color difference equations, and psychometric color terms, see Supplement No. 2 of CIE Publication No. 15, Paris.

TABLE 3			
Disease resistance scores for ‘Monterey’ and three comparison cultivars; <i>Phytophthora</i> and <i>Verticillium</i> scores were obtained in evaluations conducted in 2004-2006, <i>Colletotrichum</i> was evaluated in 2005-2006.			
Genotype	<i>Phytophthora</i> Resistance Score (5 = best)	<i>Verticillium</i> Resistance Score (5 = best)	<i>Colletotrichum</i> Resistance Score (5 = best)
‘Aromas’	4.0	4.5	2.4
‘Diamante’	2.0	2.8	2.6
‘Albion’	4.3	3.8	3.1
‘Monterey’	3.2	3.4	2.6

TABLE 4				
Flower and fruit characters for ‘Monterey’ and three comparison cultivars.				
Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
Petal number				
mean	5.5	5.4	5.6	6.5
range	5-7	5-6	5-7	5-7
Petal shape				
apex	truncate to slightly obtuse	truncate to slightly obtuse	truncate to slightly obtuse	truncate to slightly obtuse
base	attenuate	attenuate	attenuate	attenuate
margin	entire	entire	entire	entire
Petal length (mm)				
mean	10.1	9.2	9.6	10.7
range	8-11	7-13	8-11	9-13
Petal width (mm)				
mean	11.8	10.6	9.0	11.3
range	10-13	10-13	7-10	10-13
Flower position	most even	most even	most	most
(relative to foliage)	some exposed	some internal and exposed	exposed, some even	exposed, some even

TABLE 4-continued

Flower and fruit characters for ‘Monterey’ and three comparison cultivars.				
Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
<u>Calyx diam.(mm)</u>				
mean	31.3	32.0	37.5	33.2
range	28-33	25-41	31-48	29-40
<u>Corolla diam.(mm)</u>				
mean	31.2	23.9	27.8	32.9
range	26-35	18-31	23-33	25-41
<u>Sepal length (mm)</u>				
mean	12.3	12.1	14.1	11.1
range	8-15	10-15	11-18	10-12
<u>Sepal width (mm)</u>				
mean	6.4	6.7	6.6	6.8
range	3-9	5-9	4-10	5-8
Sepal color (Munsell)	7.5 GY 6/8	5 GY 5/6	2.5 GY 6/8	5 GY 6/8
<u>Pedicel length (mm)</u>				
mean	172	140	218	246
range	112-230	110-165	180-270	190-270
<u>Pedicel diameter (mm)</u>				
mean	4.4	5.3	3.1	3.0
range	4-6	4-6	2-4	2-4
Pedicel color	5 GY 6/8	5 GY 7/10	5 GY 6/8	5 GY 7/10
<u>Fruit shape</u>				
<u>Fruit length (mm)</u>				
mean	46.6	46.4	61.7	57.2
range	42-52	39-50	50-76	50-62
<u>Fruit width (mm)</u>				
mean	39.4	40.7	46.6	47.9
range	37-43	38-46	37-52	42-55
<u>Length/width</u>				
ratio	1.2	1.1	1.3	1.2
range	1.0-1.4	1.0-1.2	1.2-1.5	1.1-1.3
subjective	mostly medium to short flat conic	rounded to flat conic	most long symmetrical conic	medium conic

TABLE 4-continued

Flower and fruit characters for ‘Monterey’ and three comparison cultivars.				
Character	Cultivar			
	‘Aromas’	‘Diamante’	‘Albion’	‘Monterey’
<u>Primary/secondary fruit comparison</u>				
size	60-80%	60-80%	60-70%	60-80%
(subjective) shape	similar shape	similar shape	similar shape	similar shape
Extent/size of hollow core	small-absent	small-absent	small-medium	small-absent
<u>Calyx</u>				
position	indented-even with neck	even-indented	even-reflexed	even-reflexed
size relative to fruit	equal or greater than fruit diameter	equal or greater than fruit diameter	equal or greater than fruit diameter	equal or greater than fruit diameter
Seed position	indented-extruded	indented-even	indented-extruded	even-indented
Adherence of Calyx to Fruit	intermediate	intermediate	intermediate	intermediate

Flower measurements and fruit measurements obtained May 9–Jun. 6, 2006. Subjective observations obtained Jul. 31, 2006.

TABLE 5

Performance ‘Monterey’ and three comparison cultivars evaluated at the Watsonville Research Facility in 2005-7. All plants for these trials were harvested from a commercial nursery near Macdoel, CA on October 15-16, and transplanted after 18-21 days supplemental storage. Fruit harvest was initiated in early April and continued through the first week of October. (52" 2-row beds, 17,300 plants/acre).				
Item	Yield (g/plant)	Appearance Score (5 = best)	Fruit Size (g/fruit)	Firmness
‘Aromas’	3,108	3.1	27.0	9.6
‘Diamante’	2,653	3.5	31.2	11.0
‘Albion’	2,461	3.9	30.5	11.1
‘Monterey’	3,301	3.4	32.4	11.1

What is claimed is:
1. A new and distinct cultivar of strawberry plant having the characteristics substantially as described and illustrated herein.

* * * * *

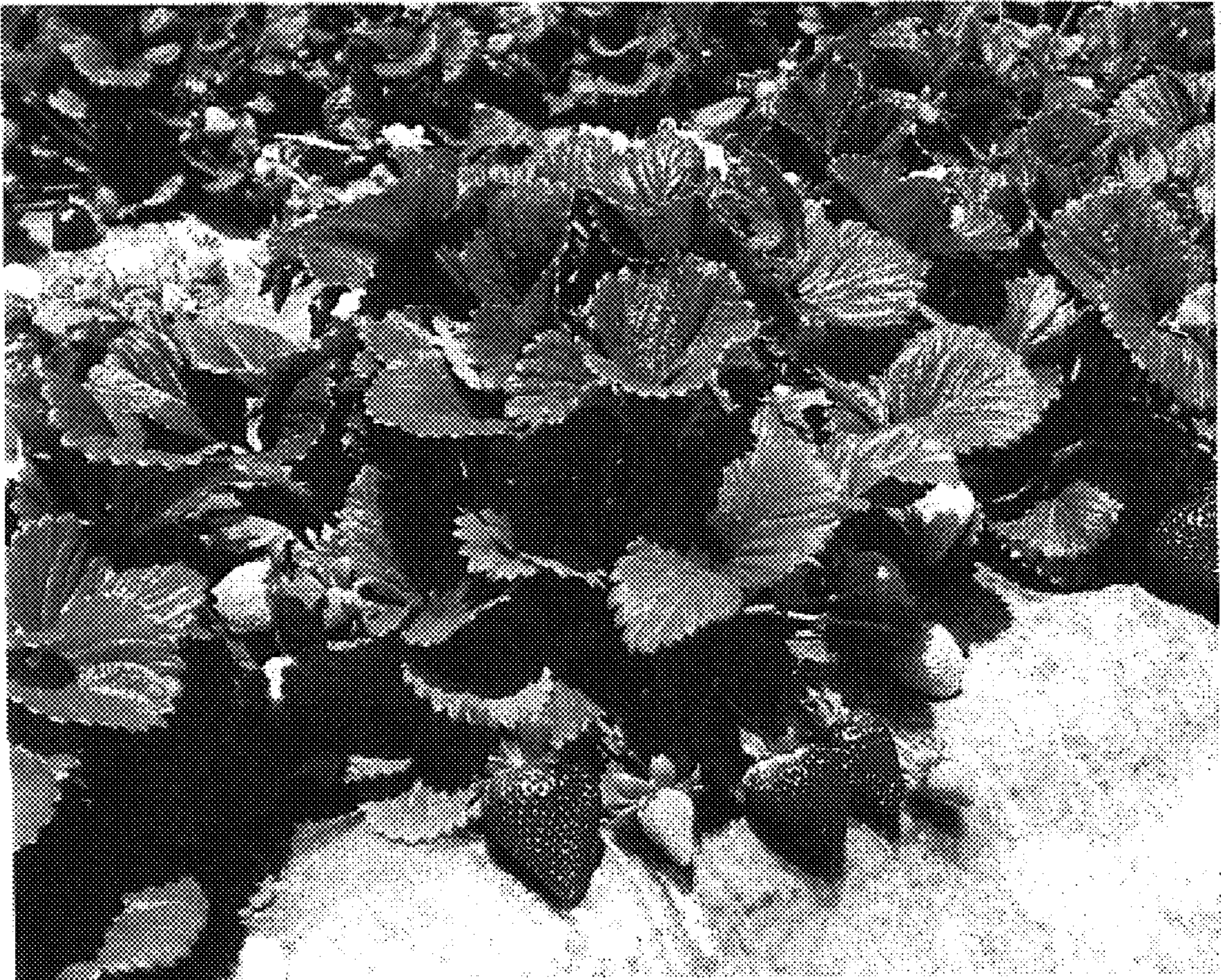


FIG. 1



FIG. 2



FIG. 3